

	(d)	acceleration is not constant.....	B1	[1]
2	(a)	 (-1 for each error or omission)	B2	[2]
	(b)	heat lost by liquid gold = $0.95m \times 129 \times \Delta T$ heat gained (silver) = $0.05m \times 235 \times (1340 - 300) + 0.05m \times 105\,000$.. $122.5m\Delta T = 17\,470m$ $\Delta T = 143\text{ K}$ temperature = $143 + 1340 = 1483\text{ K}$	C1 C1, C1 C1 A1	[5]
	(c)	e.g. thermocouple/resistance thermometer	B1	[1]
3	(a)	f_0 is at natural frequency of spring (system)	B1	